

Question	Answer	Marks
1(a)	velocity and acceleration both have constant magnitude	B1
	velocity is (always) perpendicular to acceleration	B1
1(b)(i)	$v = R\omega$	A1
1(b)(ii)	$a = R\omega^2$ or $a = v^2 / R$	C1
	$a = v\omega$	A1
1(c)(i)	$x = R \sin \theta$	A1
1(c)(ii)	$\theta = \omega t$	A1
1(c)(iii)	clear substitution of $\theta = \omega t$ into $x = R \sin \theta$ leading to $x = R \sin \omega t$	A1
1(c)(iv)	equation is of the form $x = x_0 \sin \omega t$ (so simple harmonic motion)	B1
1(d)(i)	amplitude = $0.46 / 2$ = 0.23 m	A1
1(d)(ii)	$\omega = 2\pi / T$	C1
	period = $2\pi / 1.9$ = 3.3 s	A1

Question	Answer	Marks
1(d)(iii)	$a_0 = \omega^2 x_0$	C1
	$= 1.9^2 \times 0.23$	A1
	$= 0.83 \text{ m s}^{-2}$	
1(e)	shadow on screen, labelled A, above left-hand edge of the circular path	B1

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